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Roll No 02

Implementation of a Battery-Aware Simple Reflex Vacuum Cleaner Agent

Aim

To implement a Simple Reflex Vacuum Cleaner Agent in Python that cleans different areas of a college library based on their current cleanliness condition and evaluates its performance using a scoring mechanism.

Theory

Simple Reflex Agent

A **Simple Reflex Agent** is an AI agent that selects an action based only on the **current percept** received from the environment. It does not use past experiences or maintain a history of previous states.

In this implementation, a vacuum cleaner operates in different areas of a college library:

- **Reading Hall**
- **Book Section**
- **Reception**
- **Cabin 1**
- **Cabin 2**

Each area can have one of two states:

- **0 → Clean**
- **1 → Dirty**

Condition-Action Rules

Current Percept	Action	Performance
Area is Dirty	Clean the area	+1
Area is Clean	No Action	0
Move to next area	Move	-1
Battery is Low	Go to charging station	Battery restored

The initial condition of each library area is generated randomly using Python's `random` module.

The vacuum cleaner also has a **battery level**. Cleaning and movement consume battery power. When the battery becomes low, the agent goes to the charging station and restores its battery to 100%.

The **performance score** measures how efficiently the vacuum cleaner performs its task. Cleaning provides a positive reward, while movement has a negative cost.

Components of the Agent

Environment:

Represents the college library and stores the cleanliness condition of each area.

Agent:

Represents the vacuum cleaner and decides which action to perform.

Percept:

The current cleanliness condition of the area and the battery condition.

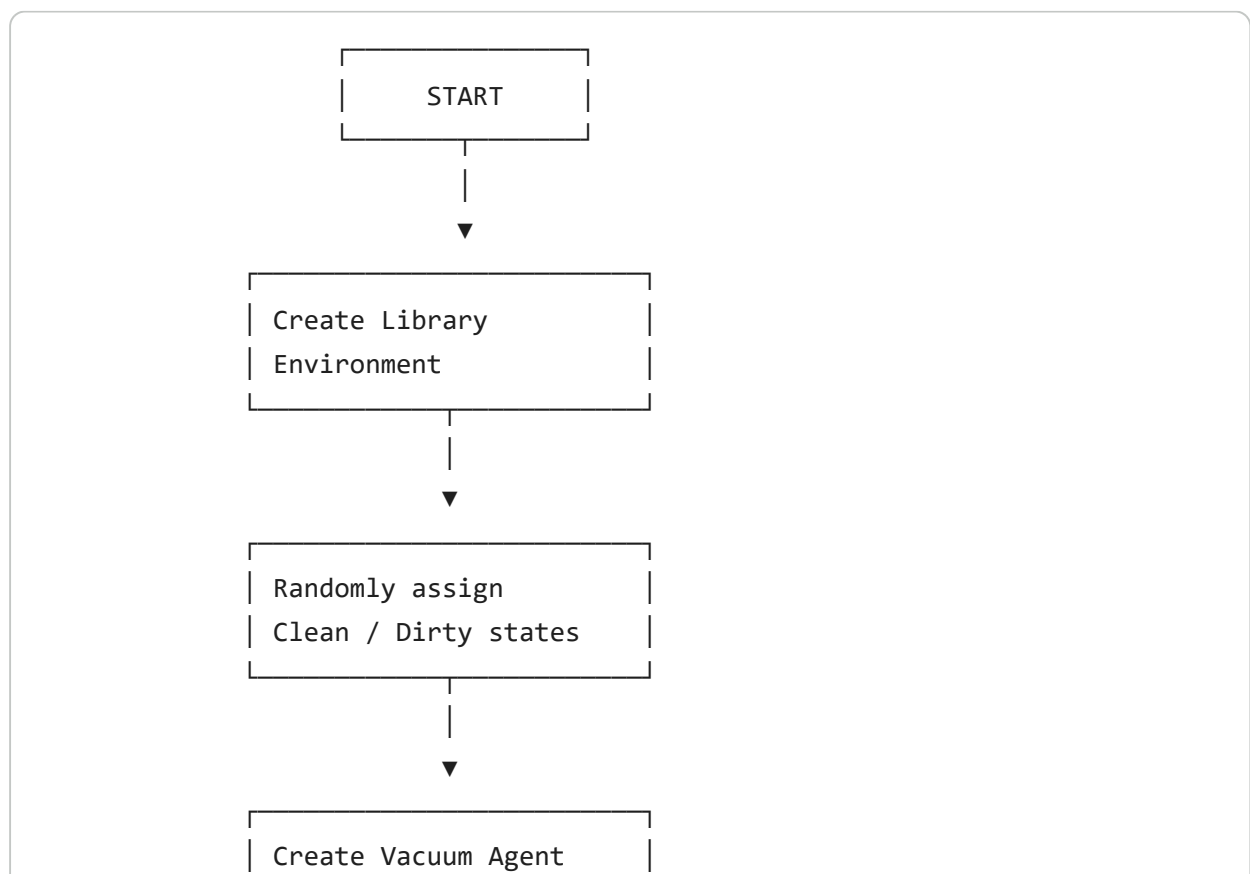
Actions:

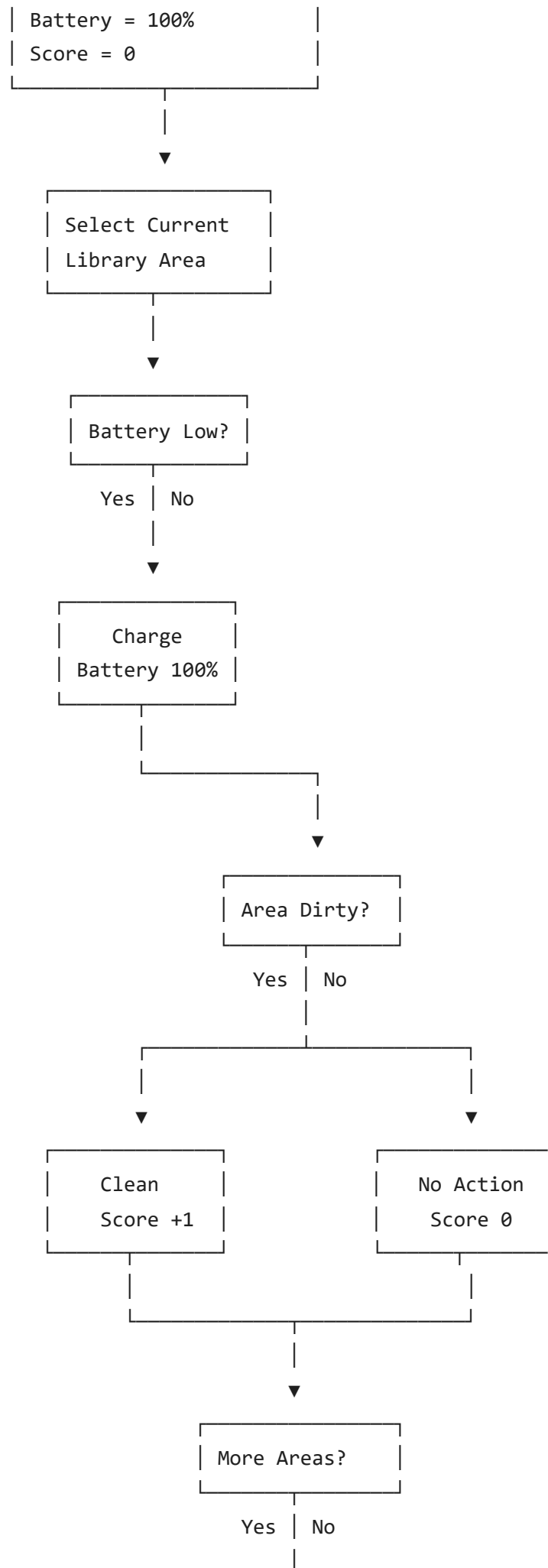
The vacuum cleaner can clean, move, or charge.

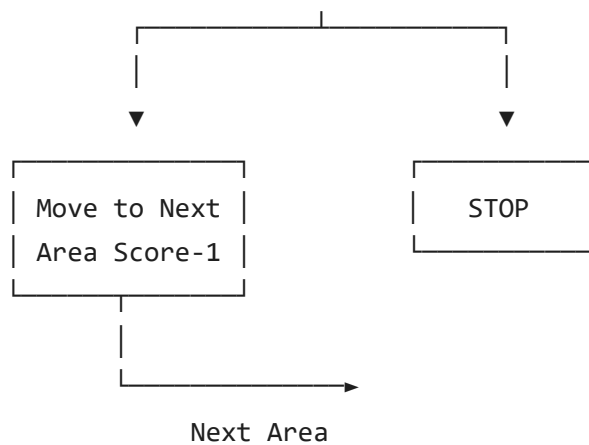
Performance Measure:

The agent receives `+1` for cleaning a dirty area and `-1` for moving to another area.

Flowchart







```
import random
```

```
# =====
# ENVIRONMENT CLASS
# Represents the MITAOE college library
# =====
```

```
class Environment:
```

```
    def __init__(self):
```

```
        # 0 = Clean
        # 1 = Dirty
```

```
        # The initial condition of every area is random.
        # This makes every run slightly different.
```

```
        self.rooms = {
            "Reading Hall": random.randint(0, 1),
            "Book Section": random.randint(0, 1),
            "Reception": random.randint(0, 1),
            "Cabin 1": random.randint(0, 1),
            "Cabin 2": random.randint(0, 1)
        }
```

```
# -----
# Returns the current condition of a particular area
# -----
```

```
def get_status(self, room):
    return self.rooms[room]
```

```
# -----
# Cleans a dirty area
# -----
```

```
def clean_room(self, room):
    self.rooms[room] = 0
```

```
# -----
# Displays the current condition of the library
```

```

# -----
def display(self):

    print("\nCurrent Library Environment:")

    for room, status in self.rooms.items():

        if status == 1:
            print(room, "-> Dirty")
        else:
            print(room, "-> Clean")

# =====
# SIMPLE REFLEX AGENT CLASS
# Represents the automatic vacuum cleaner
# =====

class ReflexAgent:

    def __init__(self, environment):

        # Connect the vacuum cleaner with the environment
        self.environment = environment

        # The vacuum starts from the Reading Hall
        self.location = "Reading Hall"

        # Initial performance score
        self.score = 0

        # Battery starts fully charged
        self.battery = 100

        # Number of areas cleaned by the vacuum
        self.cleaned_rooms = 0

        # Charging station is assumed to be near Reception
        self.charging_station = "Reception"

# -----
# Charges the vacuum when its battery is low
# -----
def charge(self):

    print("\nBattery is low!")
    print("Vacuum is going to the charging station...")

    # Move the vacuum to the charging station
    self.location = self.charging_station

    # Restore battery
    self.battery = 100

    print("Battery fully charged: 100%")

```

```

# -----
# Main decision-making method of the reflex agent
# -----
def act(self):

    # Order in which the vacuum visits the library
    locations = [
        "Reading Hall",
        "Book Section",
        "Reception",
        "Cabin 1",
        "Cabin 2"
    ]

    # Visit each area one by one
    for i in range(len(locations)):

        # Update the current location
        self.location = locations[i]

        print("\n-----")
        print("Vacuum is at:", self.location)
        print("Battery:", self.battery, "%")
        print("Score:", self.score)

        # -----
        # Check battery before performing any action
        # -----

        if self.battery <= 20:

            self.charge()

        # -----
        # Check whether the current area is dirty
        # -----

        status = self.environment.get_status(self.location)

        # If the area is dirty
        if status == 1:

            print("Percept: Area is Dirty")
            print("Action: Cleaning the area...")

            # Clean the area
            self.environment.clean_room(self.location)

            # Cleaning gives a reward
            self.score += 1

            # Cleaning consumes battery
            self.battery -= 2

```

```

        # Increase number of cleaned areas
        self.cleaned_rooms += 1

        print("Area cleaned successfully.")

    # If the area is already clean
    else:

        print("Percept: Area is Clean")
        print("Action: No cleaning required.")

    # -----
    # Move to the next area
    # -----

    if i < len(locations) - 1:

        print("Moving to the next area...")

        # Movement has a cost
        self.score -= 1

        # Movement consumes battery

        self.battery -= 1

    # FINAL RESULT

    print("\n=====")
    print("      CLEANING COMPLETED")
    print("=====")

    print("\nFinal Library Condition:")
    self.environment.display()

    print("\n----- Agent Performance -----")
    print("Areas cleaned:", self.cleaned_rooms)
    print("Remaining battery:", self.battery, "%")
    print("Final performance score:", self.score)

# MAIN PROGRAM

# Create the library environment
environment = Environment()

# Show the initial condition of the library
print("=====")
print("      MITAOE LIBRARY ENVIRONMENT")
print("=====")

environment.display()

# Create the vacuum cleaner agent

```

```

agent = ReflexAgent(environment)

# Start the vacuum cleaner
print("\n=====")
print("          VACUUM AGENT STARTS")
print("=====")

agent.act()

```

```

=====
          MITAOE LIBRARY ENVIRONMENT
=====

```

```

Current Library Environment:
Reading Hall -> Clean
Book Section -> Clean
Reception -> Clean
Cabin 1 -> Clean
Cabin 2 -> Dirty

```

```

=====
          VACUUM AGENT STARTS
=====

```

```

-----
Vacuum is at: Reading Hall
Battery: 100 %
Score: 0
Percept: Area is Clean
Action: No cleaning required.
Moving to the next area...

```

```

-----
Vacuum is at: Book Section
Battery: 99 %
Score: -1
Percept: Area is Clean
Action: No cleaning required.
Moving to the next area...

```

```

-----
Vacuum is at: Reception
Battery: 98 %
Score: -2
Percept: Area is Clean
Action: No cleaning required.
Moving to the next area...

```

```

-----
Vacuum is at: Cabin 1
Battery: 97 %
Score: -3
Percept: Area is Clean
Action: No cleaning required.
Moving to the next area...

```

```

-----
Vacuum is at: Cabin 2
Battery: 96 %

```



```
Score: -4  
Percept: Area is Dirty  
Action: Cleaning the area...  
Area cleaned successfully.
```

```
=====  
      CLEANING COMPLETED
```